Exercice 1 :

using the MME :

- μ is estimated by the sample mean :

$$\hat{\mu} = \frac{1}{5} \sum_{t=1}^{5} y_t$$

$$= \frac{6+5+4+6+4}{5}$$

$$\boxed{\hat{\mu} = 5}$$

- Similarly the MME of γ_0 is the sample variance:

$$\hat{\gamma}_0 = E[(y_t - \hat{\mu})^2] = \frac{1}{5-1} \sum_{t=1}^{5} (y_t - \hat{\mu})^2 \quad \text{(unbiased estimator)}$$

$$= \frac{1}{4} (1^2 + 0 + 1^2 + 1^2 + 1^2)$$

$$\boxed{\hat{\gamma}_0 = 1}$$

- finally the MME of ℓ_1 is the sample autocorrelation

$$\hat{\ell}_1 = r_1 = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{1}{\hat{\gamma}_0} \frac{1}{5} \sum_{t=2}^{5} (y_t - \hat{\mu})(y_{t-1} - \hat{\mu})$$

$$\hat{\rho}_1 = \frac{1}{5}\left(0 + 0 + (-1)\cdot(1) + (1)\cdot(-1)\right)$$

$$\boxed{\hat{\rho}_1 = -\frac{2}{5}}$$